

Nitrogen

What is the Role of Nitrogen (N) in Plants?

Nitrogen is an essential plant nutrient being a component of amino acids, nucleic acids, nucleotides, chlorophyll, enzymes, and hormones. N promotes rapid plant growth and improves grain yield and grain quality through higher tillering, leaf area development, grain formation, grain filling, and protein synthesis. N is highly mobile within the plant and soil.

Why Apply N Fertilizers to Rice?

Nitrogen is the most limiting element in almost all soils. Thus, proper application of N fertilizers is vital to improve crop growth and grain yields, especially in intensive agricultural systems. Insufficient and/or inappropriate fertilizer N management can be detrimental to crops and the environment. Optimal N management strategies aim at matching fertilizer N supply with actual crop demand, thus maximizing crop N uptake and reducing N losses to the environment.

How to Manage N?

- **N deficiency symptoms.** Stunted plants with small yellowish green leaves and fewer tillers; lower yield due to less panicles per unit area and less grains per panicle.
- **Occurrence of N deficiency.** Almost all soils are deficient in N; coarse textured acid soils with low organic matter content (less than 0.5% organic C); acid sulfate, saline, poorly drained, and P-deficient soils with low N mineralization and biological N fixation capacity; calcareous and alkaline soils with low soil organic matter level and a high potential for ammonia volatilization.
- **How much N to apply?** Inorganic fertilizer is the common source of supplemental N, and it is typically more profitable for farmers than purchased organic N fertilizers. Organic N sources available on-farm such as farmyard manure and compost can be effective and financially attractive in partially meeting the N requirement of rice. Apply 40-50 kg N ha⁻¹ as inorganic fertilizer for every ton of additional grain yield over yield without N. At optimum levels of nutrition, the rice crop (straw plus grain) takes up around 16 kg N per ton of grain yield (10 kg N in grain + 6 kg N in straw).
- **When to Apply N Fertilizers?** Leaf color and crop appearance indicate the plant N status and help determine the need for N fertilizer application. See Fact Sheets on i) Leaf Color Chart (LCC) for crop need-based N management, and ii) Nitrogen split applications for growth stage-based N management using the LCC.

What are the Sources of N?

- Ammonium sulfate (21% N, 24% S)
- Urea (46% N)
- Diammonium phosphate or DAP (18% N; 44-46% P₂O₅).



Applying N fertilizer (Urea) during the growing season. Photos: Dobermann & Fairburn (2000).



Nitrogen deficient rice plants on left compared with plants with sufficient N.

Further information

Fairhurst T, Witt C. editors. 2002. Rice: A practical guide to nutrient management. Singapore and Makati City: Potash & Phosphate Institute, Potash & Phosphate Institute of Canada (PPIC) and International Rice Research Institute (IRRI). p 1-89.

For information on site-specific nutrient management, visit <http://www.knowledgebank.irri.org/ssnm>.

For keys to diagnose problems in the field, visit <http://www.knowledgebank.irri.org/ricedoctor>.

For an overall view of crop management practices, visit <http://www.knowledgebank.irri.org/tropRice>.

Developed with input from V Balasubramanian and M Bell.

Phosphorus (P)

What is the Role of Phosphorus (P) in Plants?

Phosphorus is an essential plant nutrient important for root development, tillering, early flowering, and ripening. It is mobile within the plant, but not in the soil.

How to Manage P?

- **P deficiency symptoms.** Stunted dark green plants with erect leaves and reduced tillering; thin and spindly stems; delayed maturity (and no flowering at all with severe P deficiency); and high levels of unfilled grains
- **Occurrence of P deficiency.** P is often deficient in sandy soils with low organic matter content; calcareous/saline/alkaline soils; degraded lowland soils; volcanic ash soils or acid upland soils with high P fixation capacity; peat soils; and, acid sulfate soils high in active iron and aluminum.
- **How much P to apply?** At optimum plant nutrition, the rice crop (straw plus grain) takes up around 6.4 kg P_2O_5 (2.8 kg P) per ton of grain yield (4.4 kg P_2O_5 in grain and 2.0 kg P_2O_5 in straw). Recommendations for P are based on yield goal and soil P status (see Table on opposite page) as determined by grain yield in P-omission plots (see also Fact Sheet on Nutrient Omission Plot Technique for P and K).
- **When to apply P fertilizers?** Incorporate all fertilizer P before the last soil puddling before transplanting or topdress all P within 10-15 days after direct seeding.

What are the Sources of P?

P fertilizer sources and fertilizer P_2O_5 equivalents.

		Fertilizer P ₂ O ₅ (kg ha ⁻¹)				
P fertilizer	% P ₂ O ₅	15	20	30	40	60
		Amount of fertilizer required (kg ha ⁻¹)				
Single super	16-18	88	117	176	234	352
Double super/SP36	36	42	56	84	112	168
Triple super	44-46	33	44	66	88	132
Diammonium phosphate (DAP)*	44-46	33	44	66	88	**

* Also contains 18% N. ** At high P_2O_5 rates, combine DAP with other P fertilizers to prevent over application of basal N. **Note:** 1 kg P_2O_5 = 0.44 kg P and 1 kg P = 2.29 kg P_2O_5

Yield target in t ha ⁻¹ →		4	5	6	7	8
Soil P status	Yield in 0 P plot (t ha ⁻¹)	Recommended fertilizer P_2O_5 in kg ha ⁻¹				
Low	3	20	40	60	◀	◀
	4	15	25	40	60	◀
Medium	5	-	20	30	40	60
	6	-	-	25	35	45
High	7	-	-	-	30	40
	8	-	-	-	-	35

Fertilizer P_2O_5 recommendations based on yield targets and P-limited yield in P-omission plots (zero P plots). ◀ indicates possibly unrealistic yield goal.

For more information:

To learn about site-specific nutrient management, visit <http://www.knowledgebank.irri.org/ssnm>.

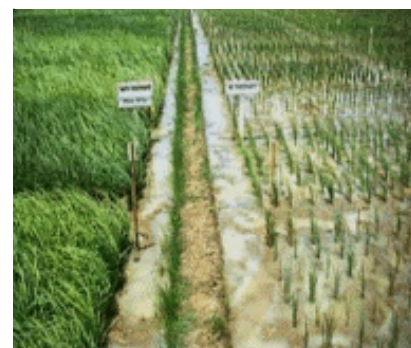
To access a key to diagnose problems in the field, visit <http://www.knowledgebank.irri.org/ricedoctor>.

For an overall view of crop management practices, visit <http://www.knowledgebank.irri.org/tropRice>.

Developed with input from V Balasubramanian, C. Witt, RJ Buresh, and M Bell.



P deficient plants are stunted, and have erect leaves compared with normal plants.



Tillering is reduced in P deficient crops. Photos: Dobermann & Fairhurst (2000).



Leaf discoloration is common to P deficient plants.

Potassium (K)

What is the Role of Potassium (K) in Plants?

Potassium is an essential plant nutrient that improves root growth and plant vigor, helps prevent lodging and enhances crop resistance to pests and diseases. Potassium is mobile in the plant and quite mobile in the soil.

Why Apply Potassium to Rice?

Potassium is often the most limiting nutrient after nitrogen (N) in high yielding rice systems. Potassium fertilizer needs to be applied in adequate amounts in most irrigated rice fields. Other nutrients need to be applied in balanced amounts to ensure a good crop response to fertilizer K application and to achieve a healthy and productive crop.

How to Manage Potassium?

- **K deficiency symptoms.** Stunted dark green plants with yellowish brown leaf margins and/or older leaves with necrotic tips and margins; leaf symptoms of K deficiency can be confused with that of tungro disease, but tungro occurs in patches in a field (not in the whole field) and usually has more pronounced yellow and orange leaves and plant stunting; leaf symptoms often appear in late growth stages; unhealthy or black roots; greater lodging; higher level of unfilled grains; lower grain weight.
- **Occurrence of K deficiency.** K deficiency occurs in intensively cropped areas with high levels of N and P application. K is often deficient in coarse-textured/sandy soils; acid upland soils; degraded lowland soils; acid sulfate soils; and, organic soils. Note: additions of K from irrigation water can be significant in some areas (e.g. South Vietnam).
- **How much K to apply?** At optimum plant nutrition, the rice crop (straw plus in grain) takes up around 19 kg K₂O (16 kg K) per ton of grain yield (2.2 kg K₂O in grain and 16.8 kg K₂O in straw). Recommendations for K are based on yield target and soil K status (see Table on opposite page) as determined by grain yield in K-omission plots (see also Fact Sheet on Nutrient Omission Plot Technique for P and K).
- **When to apply K fertilizer?** If fertilizer K rates are small, incorporate all fertilizer K before the last soil puddling before transplanting or topdress all K within 10-15 days after direct seeding. At rates > 30 kg K₂O/ha, apply 50% basal and 50% at early panicle initiation. Split K in at least two doses if soil is sandy with leaching. Use of K at flowering increases resistance to lodging and diseases in dense canopies with high yield target, but may not increase yields.

What are the Sources of Potassium?

The most common K fertilizer is muriate of potash (MOP) which contains about 50% K or 60% K₂O in the form of KCl (30 kg K₂O is equal to 50 kg MOP or KCl). Rice straw is rich in K (14.0 kg K or 16.8 kg K₂O per ton straw). Note: 1 kg K₂O = 0.83 kg K and 1 kg K = 1.2 kg K₂O

Fertilizer K₂O recommendations based on yield targets and K-limited yield in K-omission plots (zero K plots) at medium levels of straw return (2-3 t/ha).

Yield target in t ha ⁻¹ →		4	5	6	7	8
Soil K status	Yield in 0 K plot (t ha ⁻¹)	Recommended fertilizer K ₂ O in kg ha ⁻¹				
Low	3	30	60	90	◀	◀
	4	0	35	65	95	◀
Medium	5	-	20*	50*	80*	110*
	6	-	-	35*	65*	95*
High	7	-	-	-	50*	80*
	8	-	-	-	-	65*

◀ indicates possibly unrealistic yield goal.

* Lower K₂O rates by about 20-25 kg K₂O/ha, if 4-5 t straw/ha are returned after harvest, if K rich sediment inputs are high, or if long-term experiments indicate high soil K supplying power. Increase K₂O rates by the same amount, if straw is fully removed after harvest.

Further information:

For information on site-specific nutrient management, visit <http://www.knowledgebank.irri.org/ssnm>.
For a key to diagnose problems in the field, visit RiceDoctor at <http://www.knowledgebank.irri.org/ricedoctor>.
For an overall view of crop management practices, visit Troprice at <http://www.knowledgebank.irri.org/troprice>.

Developed with inputs from V Balasubramanian, C Witt, RJ Buresh, and M Bell.

Sulfur (S)

What is the Role of Sulfur (S) in Plants?

Sulfur is an essential plant nutrient required for chlorophyll production. It is needed to produce essential amino acids (cysteine, methionine, and cystine) in plants that are related to human nutrition. S is quite mobile in the plant (though less so than N) but only partially mobile in the soil.

Why Apply Sulfur Fertilizer to Rice?

Sulfur limits plant growth when the soil supply of S is low or adverse soil conditions prevent plant uptake of S. In such cases S needs to be applied as required. Other nutrients need to be applied in balanced amounts to ensure a good crop response to fertilizer S application and to achieve a healthy and productive crop.

How to Manage Sulfur?

- **S deficiency symptoms.** Pale green plants; pronounced pale yellowing of young leaves (in contrast to yellowing and progressive death of older leaves for N deficiency). Soil and/or plant analysis is often needed to confirm S deficiency.
- **Occurrence of S deficiency.** S deficiencies are not very common. S may be needed in leached sandy soils; low organic matter soils; and, highly weathered soils rich in iron oxides.
- **How much S to apply?** Apply 10 kg S ha⁻¹ at moderate S deficiency and 20-40 kg S ha⁻¹ at severe deficiency. The crop requires around 2 kg S ha⁻¹ (both straw and grain) per t of grain yield.
- **When to apply S?** When required, apply any S fertilizer just before the last puddling along with P and K. The effect of applied S can last 2 crop seasons.

What are the Sources of S?

The commonly used S fertilizers are ammonium sulfate (24% S), single super phosphate (12% S), and gypsum (17% S).



Yellowing of youngest leaves in S-deficient plants. Photos: Dobermann & Fairhurst 2000.



Leaf canopy in S-deficient plot appears yellow.

Further reading

Dobermann A, Fairhurst T. 2000. Rice: Nutrient disorders and nutrient management. Singapore and Los Baños: Potash & Phosphate Institute (PPI), Potash & Phosphate Institute of Canada (PPIC), and International Rice Research Institute (IRRI). p 1-191.

For more information:

For information related to site-specific nutrient management, visit <http://www.knowledgebank.irri.org/ssnm>.

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Developed with inputs from V Balasubramanian, RJ Buresh and M Bell.

Zinc (Zn)

What is the Role of Zinc (Zn) in Plants?

Zinc is an essential plant nutrient required for several biochemical processes in the rice plant, including chlorophyll production and membrane integrity. Thus, Zn deficiencies affect plant color and turgor. Zinc is only slightly mobile in the plant and quite immobile in soil.

Why Apply Zinc to Rice?

Zinc limits plant growth when the soil supply of Zn is low or adverse soil conditions (such as continuous flooding) prevent plant uptake of Zn. In such cases, Zn needs to be applied as required. Other nutrients need to be applied in balanced amounts to ensure a good crop response to fertilizer Zn application and to achieve a healthy and productive crop.

How to Manage Zn?

- **Zn deficiency symptoms.** Stunted plants and dusty brown spots on upper leaves; patches of poorly established plants; symptoms appear 2 to 4 weeks after transplanting; higher levels of empty grains; delayed maturity and lower yields; Zn deficiency leaf symptoms resemble S and Fe deficiency in alkaline soils and iron toxicity in poorly drained organic soils.
- **Occurrence of Zn deficiency.** Zn deficiency is not very common, but can occur in neutral and calcareous soils; intensively cropped soils; continuously flooded paddy soils or very poorly drained soils; sodic and saline soils; peat soils, soils with high available P and Si status; sandy soils; highly weathered, acid, and coarse-textured soils; soils derived from serpentine and laterite; and, leached, old acid sulfate soils with a small concentration of K, Mg, and Ca.
- **How much Zn to apply?** If Zn deficiency symptoms are observed in the field, apply 10-25 kg $\text{ZnSO}_4 \cdot \text{H}_2\text{O}$ or 20-40 kg $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ per ha on the soil surface, or dip roots of rice seedlings in a 2-4% ZnO suspension before planting (i.e., 20-40 g ZnO/L water). Rice plants can recover from Zn deficiency if the field is drained - a dry fallow increases the availability of Zn. The crop only requires around 0.05 kg Zn ha^{-1} (both straw and grain) per ton of grain yield, but much more Zn fertilizer must be applied because Zn once applied is not very available to the plant.
- **When to apply Zn?** Apply the Zn fertilizer on the soil surface after last puddling and leveling in the main field or apply Zn fertilizer to the nursery beds 7-8 days before pulling seedlings. The effect of zinc application to soil can last 2 to 5 crop seasons on all soils except in alkaline soils. In alkaline soils, zinc may need to be applied to each crop.

NOTE: Aerating soil—allowing it to dry out—can reduce Zn deficiency.

What are the Sources of Zn?

The commonly used Zn fertilizers are soluble zinc sulfate (23-36% Zn), soluble zinc chloride (48-50% Zn), and insoluble zinc oxide (60-80% Zn).

Further reading

Dobermann A, Fairhurst T. 2000. Rice: Nutrient disorders and nutrient management. Singapore and Los Baños: Potash & Phosphate Institute (PPI), Potash & Phosphate Institute of Canada (PPIC), and International Rice Research Institute (IRRI). p 1-191.

For more information:

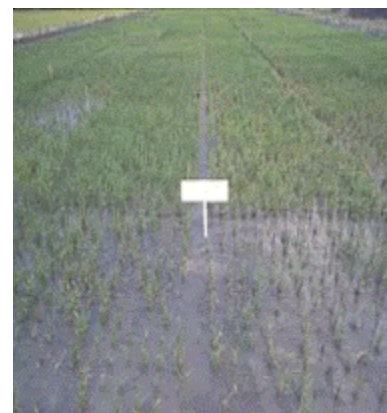
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Appearance of dusty brown spots on upper leaves of Zn-deficient plant.



Uneven field growth in a Zn-deficient plot. Dobermann & Fairhurst (2000).